Chapter # 01 Physical Quantities

Convert the following numbers in Standard form / scientific notation.

a) 149,530,000,000 m which is the average distance between earth and Sun.

b) 0.0008 g which is the average mass of human hair.

c) The number of seconds in a day.

What are prefixes? What is their use in measurements?

How many kilometers are there in 25 micrometers?

Why do we need to consider significant digits in measurements?

How random error can be reduced?

Differentiate between precision and accuracy.

Express the followings measurements using prefixes

a. 27.5×10^{-10} m b. 0.00023×10^{-2} s

Write down the rules of Significant figures and give two examples of each.

Differentiate between Scalars and Vectors with Examples

Differentiate between systematic and random error?

Write down the rules of rounding for whole numbers and decimal numbers.

Find the number of significant figures in each of the following values. Also express them in scientific notations.

a) 100.8 s b) 0.00580 km c) 210.0 g

Round off the following numbers to

(a)	Two decimal points	i) 3.876	ii) 657.873	iii) 0.0857
(b)	Three significant digits	i) 24.68	ii) 0.07683	iii) 7,847

Chapter # 02 Kinematics

Write three types of translatory motion, explain briefly with two examples each.

Why distance is a scalar while displacement is a vector quantity, justify your answer with vector diagram?

Differentiate between speed and velocity?

Differentiate between average speed and instantaneous speed?

Differentiate between uniform and variable speed? Also drawn distance time graph for both, in case of variable speed , graph must be drawn for a body moving with a) increasing variable speed b) decreasing variable speed.

Differentiate between uniform and variable velocity? Also give daily life examples of both.

Differentiate between uniform and non-uniform acceleration? Also give daily life examples of both.

Shanghai's Maglev, the fastest train, travelled a distance of 30 kilometres in 7 minutes and 30 seconds. What is its speed? Convert the speed to km/h.

A car travels a curvy track of length 800 metres in 40 seconds. The straight path is about 600 metres between starting point and ending point, which the same car travels in 36 seconds. What is the car's (a) average speed and (b) average velocity?

A boy moves for some time, give two situations in which his displacement is zero but covered distance is not zero?

A stone tied to string is whirling in circle, what is direction of its velocity at any instant?

Is it possible to accelerate an object without speeding it up or slowing it down?

Can a car moving towards right have direction of acceleration towards left?

With the help of daily life examples, describe the situations in which:

a. acceleration is in the direction of motion.

b. acceleration is against the direction of motion.

c. acceleration is zero and body is in motion.

Which controls in the car can produce acceleration or deceleration in it?

A car travels a curvy track of length 800 metres in 40 seconds. The straight path is about 600 metres between starting point and ending point, which the same car travels in 36 seconds. What is the car's (a) average speed and (b) average velocity?

What is a speed time graph and why do we use it? Draw speed time graph for a body moving with

a) uniform velocity

b) uniform acceleration

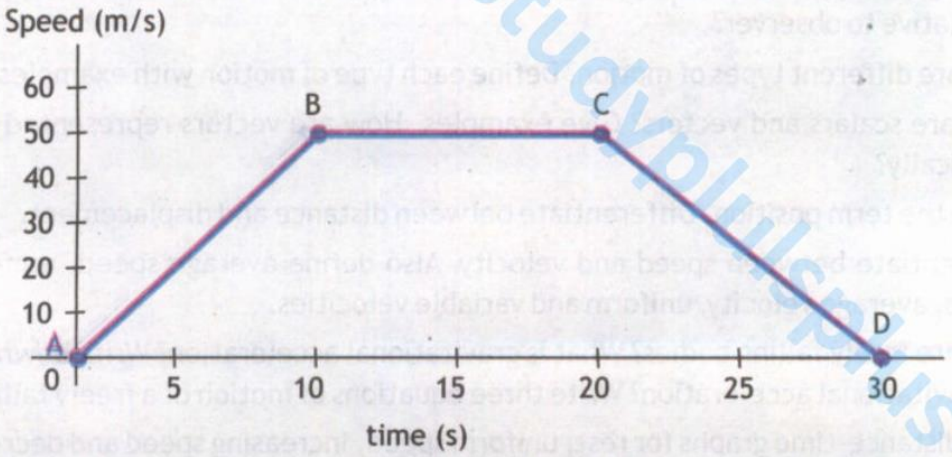
Find the gradient and distance travelled in both cases.

6. Consider the following speed time graph. Tell:

a. Which part of the graph is showing acceleration, deceleration and zero acceleration?

b. Calculate covered distance from 10 seconds to 20 seconds from the graph.

(Ans. (a) OA, BC, AB, (b) 500 m)



SPEED - TIME GRAPH

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Chapter # 03 Dynamics-I

Differentiate between contact forces and non-contact forces. Also give 3 examples of each.

Explain briefly all 4 fundamental forces in nature. Give at least two examples of each.

Differentiate between system diagram and free-body diagram? Make system and free-body diagrams for the following cases:

a) a football falling to the ground with uniform velocity
b) A book lying on a table

Why is it more difficult to push large man on swing compared to a small child?

Explain Newton's 1st Law of Motion? Give mathematical expressions and two examples.

What is inertia, explain briefly? Do all bodies have same inertia?

Explain Newton's 2nd Law of Motion? Give mathematical expression. Give two examples from daily life.

If the same engine is installed in a bus and car that applies a force of 3000 N. What acceleration will this engine produce in a bus of mass 12,000 kg and a car of mass 1200 kg?

Explain Newton's 3rd Law of Motion? Give mathematical expression. Give two examples from daily life.

Write down the limits of Newton Laws?

Differentiate between mass and weight? How do we measure both?

Mass of your school bag is 8 kg. How much will it weigh (a) here on Earth and on (b) the surface of moon? [Take acceleration due to gravity for Earth as $g_E = 9.8\ m/s^2$ and for Moon as $g_M = 1.625\ m/s^2$]

What is gravitational field? Explain briefly what gravitational field strength is , its formula , units . Does it same everywhere in the universe?

A box weighs 400 N on earth while 150 N on an unknown planet. Find the gravitational field strength on that planet.

What is Momentum and what it is used for? Write its mathematical expression and units.

A golfer hits a ball having mass 45 g. If after the shot, the ball travels with a speed of 80 m/s, what magnitude of momentum does the golfer imparted to ball?

Show that net force equals the rate of change of momentum?

How a fielder should catch the ball so that his hands will be hurtless?

What is difficult to stop if their brakes fail and are travelling from an inclined road:

(a) A car of mass 1200 kg moving with a velocity of 8 m/s in 5 seconds,

(b) A truck of mass 10,000 kg moving with the same velocity in the same time?

What is impulse and how it is related to momentum? What is the use of impulse in daily life? How do we measure the impulse?

Define an isolated system? What is law of conservation of momentum. Derive the expression of the velocity of the Gun V_G when it fires a bullet.

Ahmad fired a bullet of mass 17 g from his hunting gun, of mass 3 kg. If the velocity of the bullet were 350 m/s, what would be the recoil velocity of the air gun?

When a motor cyclist hit a stationary car, he may fly off the motor cycle and driver in the car may get neck injury. Explain

In autumn, when you shake a branch, the leaves are detached. Why?

Why it is not safe to apply brakes only on the front wheel of a bicycle?

A man slips on the oily floor; he wants to move out of this area. He is alone. He throws his bag to move out of this slippery area. Why is it so?

Why are cricket batter gloves padded with foam?

Where will your weight be greater, near earth or near moon? What about mass?

Chapter # 04 Dynamics-II

Define moment of force? Write its expression and units? Give three daily examples where do we use this principle.

Differentiate between center of mass (CM) and center of gravity (CG). What is CM/CG of different shaped objects?

Define equilibrium? Briefly explain the necessary condition for equilibrium?

What is Principle of Moments? What are its applications?

Briefly explain the types of equilibrium. Give two examples of each type.

What is stability and how is it related to CM/CG? What are equilibrium types based on stability? Give two examples of each.

How can we improve the stability of an object?

Define friction and how do we measure it, give two daily examples of friction?

Do you list any three advantages and disadvantages of friction?

How can we reduce friction? Write three methods.

Why rolling friction is less than static friction?

Explain briefly what fluid friction is and how it is related to the drag force.

Do wider tyres of a car increase or decrease friction and what is its relationship with friction?

List the factors on which drag force depend?

What is terminal velocity and how it is achieved by an object?

What is frictional dissipation? Explain it with example of sky diver and with shooting stars?

In winter , we feel warmth when we shake hands, why?

What is Centripetal force and why is it necessary? Write its equation. Which sources provide this force, give three daily life examples.

What is a satellite? Differentiate between natural and orbital satellite? How do we put a satellite into orbit?

Derive the expression for average orbital speed a satellite.

Two satellites are following one another in the same circular orbit. If one satellite tries to catch another (leading one) satellite, can it be done by increasing its speed?

Earth completes one revolution around the sun in 365.25 days. Find the orbital speed of Earth around the sun if it is 150 million km away from the sun.

Why long spanner is used to open or tight nuts of vehicle's tyre? While tightening a small nut, extra-long wrench is not suitable. Why?

Why door knobs are fixed at the edge of door? What will happen it the door knob is at the middle of the door?

If you drop a feather and a bowling ball from the same height, which one will reach the terminal velocity first? Which one of them will hit the ground first?

Why do ice skates effortlessly slide on ice, while your shoes cause skidding?

Why is a pencil standing on its tip unstable, and what factors affect the stability of an object balanced on a point?

While driving what happens if the driver take the curve too fast? How does centripetal force play a role in keeping the car from skidding off the road?

Why is it important for communication satellites in geostationary orbit to maintain a specific speed?

A car weighing 1200 kg enters a roundabout with a diameter of 60 meters at a speed of 25 km/h. Calculate the centripetal force acting on the car as it navigates the curve.

(Ans. 693.3 N)

A geostationary satellite revolves around earth in an orbit of radius 42000 km. Find orbital speed of the satellite at this height.

Chapter # 05 Pressure & Deformation

Define elasticity? Give example of daily life objects which are more elastic than others?

Define and explain Hook's Law? Write its mathematical equation? What is spring constant and what information do we get from it? Briefly explain what the limit of proportionality is.

Briefly explain the following applications of Hook's Law : Balance wheel of the mechanical watch, Spring scale and Galvanometer?

Briefly explain the stress curve. Relate following terms on the graph: proportional limit, elastic limit, yield strength, and ultimate stress. Explain briefly about elastic and plastic region.

Kamil sits on a spring chair as shown in the figure. If Kamil's weight is 50 kg and compresses the spring by about 10 cm, when he sits on the chair, find the spring constant of this chair's spring.

Define pressure, write its expression and in which units can we measure it.

Why does a balloon burst easily when pricked with a pin but not when squeezed by our hands?

Why blunt knife cannot cut meat easily like sharp knife does?

Abdurrahman was standing at the stage of a hall for a speech. What pressure does he apply on the stage if his two feet cover an area of 400 cm^2? (b). If for a while he stands on one foot, what will be the pressure under that foot? (Take his mass 50 kg).

What is atmospheric pressure ? Write its units. Do Earh and Moon have same atmospheric pressure?

Differentiate between barometer and manometer.

List three applications of liquid barometers.

Briefly explain the following applications of atmospheric pressure: drinking by a straw , drawing liquid by a syringe.

Why it is difficult to cook food at high altitudes?

Derive the relationship of Liquid pressure. What factors it depends?

A submarine was moving in the Pacific Ocean (the largest and deepest ocean) at a depth of 8.5 km. How much pressure is exerted upon the submarine if density of water is 1000 kg/m^3?

Write any 5 applications of Manometer.

What is Pascal's law? Explain its following two applications : Hydraulic Lift , Hydraulic Car Brake System

A hydraulic lift has 0.002 m^2 narrow cylinder area while 0.9 m^2 wider cylinder area. How much force must be applied at the narrow cylinder if a car weighing 1800 kg is to be lifted?

If a liquid has density twice the density of mercury, what will be height of liquid column in barometer?

Why we wouldn't be able to sip water with a straw on the moon?

How are we able to break a metal wire by bending it repeatedly?

Why do static fluids always exert a force perpendicular to the surface?

How can a small car lifter lifts load heavier than itself?

Consider a spring with a spring constant of 8000 N/m. If a force of 500 N is applied to the spring, what will be the displacement of the spring?

A hydraulic car lift lifts a car of mass 1000 kg when we apply force of 50 N on small piston. Radius of its small piston is 20 cm. Find the radius of its large piston.

How much force should be applied on an area of 20 cm^2 to get a pressure of 4500 Pa?

Water column in a beaker is 70 cm. Find the pressure of water in beaker. Take density of water as 1000 kg/m^3.

Chapter # 06 Work and Energy

Define work, write its mathematical form and units. Why work is scalar quantity while moment of force is a vector?

A man is pushing the truck but truck is at rest, is he doing work? Explain.

Muhmmad Tuaha was pushing a box with a force of 100 N on a flat frictionless surface. How much work he does on the box if he pushes it through a displacement of 200 m?

Define energy, write its unit? How it is different from work although they have same units?

Briefly explain the forms of energy.

What is Kinetic energy, derive its formula. List two examples of daily life of Kinetic energy.

Babar Azam hits a cover drive by giving kinetic energy of 50 J to the ball by his bat. At what speed will the ball go to the boundary if mass of the ball is 120 g?

What is Potential energy, derive its formula. List two examples of daily life of potential energy.

At School sports gala, Kamil made a record in high jump at his school. He jumped 1.5 m high. What will be his potential energy if his mass is 60 kg? From where does this potential energy come?

What is law of conservation of energy? Derive its mathematical expression if a ball is dropped from the roof of a building. Now includes the effect of friction in the expression.

A player kicks the football of mass 450 g is thrown vertically upward at a speed of 22 ms^{-1}.
(a) Neglecting air resistance how high would it reach. (b) If due to air friction the ball only rises to 20 m, what is the work done against air resistance?

Differentiate between renewable energy and non-renewable energy? Give examples of both.

What are fossil fuels? Write a general chemical equation of combustion of methane. Write any three important fossil fuels?

Briefly describe nuclear fuel. Give its example.

Briefly describe the bio-mass fuel? Give examples. What are best uses of biomass fuels?

What are hydroelectric resources of energy? How this energy is produced, briefly explain.

What are solar radiations and how do we get energy from them?

Briefly explain the following resources of energy: geothermal , wind, wave in the sea, and tides.

Define Power. Write its equation and units?

Show that 1KHW = 3.6 x10^6 J

How powerful will be a Tesla Car's engine if it does 3×10^7 J of work in 1 minute? What will be its power in hp?

What is efficiency and why do we measure efficiency devices like electric generators?

A petrol engine in-takes 400 J of fuel energy and does only 120 J of useful work. How efficient is this petrol engine?

A car is moving with a constant speed along a straight road. Is there any work done on the car?

A car has Kinetic energy 'E_K'. By what factor its kinetic energy would change, if its velocity is doubled?

A bullet is fired from gun, bullet penetrates into sand wall and it stops. Where does its kinetic energy used?

An LED light bulb has efficiency of 80%. Does it violate conservation of energy principle?

Will we eventually rely entirely on renewable energy sources? Why or why not?

How can increasing the power of a machine impact its energy consumption?

. A perpetual engine has an efficiency equal to 1. Why it will not work?

(Ans: 750 J)

A boy weighing 75 N jumps up and gains 300 J of gravitational potential energy. Find the height to which the boy will rise.

A 2.0 kg rock is dropped from 20 m tall building. What is the Kinetic and gravitational potential energy when the rock has fallen 15 m.

(Ans: (a) 2250 J, (b) 980 J and (c) 22.8 %

What horsepower (hp) is required to pump up 2500 kg of water to 100 m height in 5 minutes?

Chapter # 07 Density and Temperature

Define density, write its formula and units? State the densities of solids, liquids, and gasses?

You find a material in the shape of cube of side length 5 centimetre. Mass of this cube is 500 grams. Find the density of this material?

Compare the properties of solids, liquids, and gasses?

Why Pasma is known as fourth state of matter? What are the sources of plasma and write its three applications?

How does temperature affect the motion of particles? Also draw graph between temperature and K.E. of particles, and Temperature vs Pressure of particles?

What is internal energy and how it is related to temperature of a substance?

Briefly describe diverse types of thermometers like Mercury , Gas Thermometer and its types like Constant Pressure and constant volume, also explain liquid crystal thermometers.

Differentiate between Thermistors and Thermocouples thermometers.

Briefly explain the sensitivity, range, and linearity of Thermometers.

Now the structure of a liquid-in-glass thermometer affects its sensitivity, range, and linearity.

Why water is not used in liquid in glass thermometres?

Can we increase internal energy of a substance without increasing its temperature?

Why - 273.15°C temperature is called absolute zero? Can we achieve this temperature?

Can we increase the sensitivity of a liquid-in-glass thermometer without changing its range?

One student claims to have constructed a more sensitive liquid in glass thermometer. How can her claim be verified?

A 70cm 10cm 30cm plastic box has mass of 2500 g. Find the density of plastic.

(Ans. 0.12 g/cm^3)

Aluminum has a density of 2700 kg/m^3. Find the mass of a solid 25 cm diameter aluminum ball.

(Ans. 22 kg)

A cube of iron has a side length of 10 cm. What is volume of this cube? Mass of this iron cube is half kilogram. This cube has cavity inside it, find the volume of the cavity?

Chapter # 08 Magnetism

Briefly describe magnetic shielding.

What is induced magnetism? Briefly describe the induced magnetism methods like stroking, hammering, and heating.

What is a solenoid? How solenoid can function as a magnet? Write equation for the magnetic field of solenoid.

Does magnetic field lines always emerge from north pole? Can they move from south pole to the north pole? Justify.

What happens if a current passed from a conductor? Write the equation of magnetic field in this case. Draw a diagram to support your answer.

Differentiate between a permanent magnet and an electromagnet.

List some uses and applications of permanent magnet and electromagnet.

Differentiate between diamagnetic , paramagnetic, and ferromagnetic materials with examples.

Differentiate between magnetic and non-magnetic materials with examples.

How does the earth create its own magnetic field?

Explain the geographical and magnetic poles of the Earth?

What is meant by bio magnetism? Write down its applications.

A student takes a long straight copper wire (with insulation on it) from his physics lab, he wraps the wire on an iron rod of 50 cm, to make it solenoid of 15 turns. He connects the ends of copper wire with the battery, which provides 1.2 A current. Find the magnetic field he produces in solenoid by doing so.

Can two magnetic field lines intersect each other? Justify your answer.

A freely suspended magnet always points along north-south direction. Why?

What is the neutral zone or field free region of the magnetic field?

Is there any material which does not have any magnetic behavior? Justify your answer.

What is the geomagnetic reversal phenomenon? Explain.

Why the Earth spins about its geographical axis instead of its magnetic axis? Explain.

Why the Earth's geographical and magnetic axis are not coincident? Explain.

Find the magnetic field due to a wire at 10 cm, if 1.3 A current is passing through the wire.

Chapter # 9 Nature of Science

Explai briefly the following fields of Physics: Astrophysics, cosmology, optics, and astronomy.

Differentiate between classical, quantum and relativistic mechanics.

Differentiate between hypothesis, theory, and law with examples from physics.

Differentiate between science, technology, and engineering.